

Senior IT engineer who builds product-quality internal tooling to scale operations, with deep expertise in zero-trust architecture, macOS fleet management, and full-stack development. Proven track record turning manual IT processes into automated, self-service platforms.

PROFESSIONAL EXPERIENCE

Senior Client Platform Engineer, Unity Technologies

October 2020 – Present

- **Engineered a T3 stack application** (Next.js, TypeScript, tRPC, Drizzle) hosted in GCP k8s cluster, integrating disparate system data that automated license management, enabled remote device security controls, and provided analytical reporting dashboards for IT leadership; expanded platform to include inventory stock take workflows saving IT Operations over 30 hours per month, an automated device retirement process, and a stock forecasting module enabling data-driven procurement planning resulting in significant operational scaling beyond team headcount. Developed Model Context Protocol (MCP) server enabling natural language queries of internal data.
- **Built a standalone Slack workspace hygiene application** with a Next.js/React frontend and Python/FastAPI backend, deeply integrating with the Slack API to automate identification and remediation of stale channels, dormant users, and unused integrations — improving security posture, supporting user offboarding compliance, and significantly reducing workspace clutter and administrative overhead.
- **Implemented safe device management for 10,000+ macOS & Windows devices** through Infrastructure-as-Code and GitOps workflows
- **Architected a DevOps automation pipeline** utilizing serverless AutoPkg/Munki workflows with CI/CD integration, improving deployment reliability and security posture by automatically updating over 80 applications for over 4000 devices.
- **Enhanced security posture** by architecting Okta device trust solutions for Zero Trust authentication and spearheading the implementation of Kolide, including developing custom osquery-based security compliance checks for comprehensive device verification and reporting
- **Executed a 4,000+ device MDM migration** with custom tooling, including 1,500+ devices from acquisitions, while maintaining security consistency and optimizing end-user experience
- **Developed an intelligent update campaign system** for neglected devices, achieving 90%+ compliance with the latest OS, improving from 700 to 3,500+ devices on current release
- **Orchestrated multi-cloud infrastructure** using Terraform to provision & manage critical systems, including VDI environments in Azure and client management services in AWS and GCP, applying IaC principles for consistent, version-controlled deployments across cloud platforms

Application Administrator, Central Michigan University

December 2011 – October 2020

- **Founded and led cross-departmental Mac Users Group** focused on endpoint security standardization and best practices adoption
- **Designed and scaled macOS management infrastructure** from departmental to university-wide, managing ~2,000 devices with minimal resources
- **Built comprehensive patch management system** utilizing Munki, AutoPkg, and GitLab CI/CD for staged update rollouts

OPEN SOURCE CONTRIBUTIONS & PRESENTATIONS

- **Enterprise IT Automation Tools:** [Developed](#) and presented at multiple industry conferences (MacDevOpsYVR, PSU MacAdmins) on automation frameworks and IT efficiency solutions
 -  [Streamline Your macOS Administration with Github Actions - Jon Crain](#)
- **MunkiReport:** Core contributor and module developer, organized and delivered a full-day workshop with the creator Arjen van Bochoven. Implemented a demo site in Azure with automated deployment through Azure pipelines
- **munkiMDM:** Developed proof-of-concept integration between MicroMDM and Munki for unified open source management, which has been implemented in production by other admins
- **SwiftDialog:** Developed integration scripts and presented implementation strategies for MDM migrations and end-user communications
 -  [Customizable Dialogs for macOS: Using SwiftDialog for Better UX – Jon Crain](#)

TECHNICAL SKILLS & COMPETENCIES

- **Infrastructure & Cloud:** AWS, Azure, GCP, Terraform, GitOps workflows, CI/CD pipelines, k8s, serverless architecture
- **Software Development:** Python, FastAPI, TypeScript, Next.js, React, T3 stack (tRPC, Drizzle), Go, Swift, Bash
- **Identity & Access Management:** Okta, SAML, OAuth, SCIM, device trust, RBAC, Zero Trust architecture
- **Security & Compliance:** CrowdStrike, Kolide, Beyond Identity, osquery, vulnerability management
- **Endpoint & Device Management:** Jamf, SimpleMDM, Munki, AutoPkg, Intune, WorkspaceOne, Chef, Puppet — macOS, Windows, ChromeOS, Linux
- **AI & Automation:** Model Context Protocol (MCP), Claude API, automation framework design, self-service tooling
- **Enterprise SaaS:** Google Workspace, Slack, and broad enterprise SaaS administration and security

EDUCATION & CERTIFICATIONS

Bachelor of Applied Arts, Central Michigan University
ITIL 4 Foundation Certification – AXELOS

Major: Entrepreneurship